

2.1 The Periodic Table: Chemical Periodicity

Question Paper

Course	CIEA Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.1 The Periodic Table: Chemical Periodicity
Difficulty	Hard

Time allowed: 40

Score: /29

Percentage: /100

Question 1a

Sodium oxide and sulfur dioxide are separately added to water.

Write equations for and observations of each reaction and state the pH of any solution formed.

- Sodium oxide and water

.....

- Observations

.....

- Resulting pH

.....

- Sulfur dioxide and water

.....

- Observations

.....

- Resulting pH

.....

[6 marks]

Question 1b

A concentrated solution of the compound produced when sodium oxide is added to water is heated with silicon dioxide.

Write the ionic equation for this reaction.

[1 mark]

Question 1c

Compound **A** is an oxide of a Period 3 element. It dissolves in hot, concentrated potassium hydroxide and in nitric acid, but it is insoluble in water.

i)

State the term that is used to describe the behaviour of compound **A**.

[1]

ii)

Write equations, including state symbols, for the reactions of compound **A** with potassium hydroxide and nitric acid.

[2]

[3 marks]

Question 2a

The acid-base character of the Period 3 oxides changes from sodium to sulfur.

In terms of electronegativity of the Period 3 element, explain why the acid-base character changes.

[3 marks]

Question 2b

When phosphorus(V) oxide reacts with water it will produce phosphoric acid.

- i)
Write an equation to show the formation of phosphoric acid and include the expected pH of the resulting solution.

[2]

- ii)
Explain why the melting point of phosphorus(V) oxide is relatively low compared to most other Period 3 oxides.

[3]

[5 marks]

Question 2c

Phosphorus(V) pentoxide reacts with sodium hydroxide solution, NaOH (aq), to form a phosphate salt.

i)

Write the overall equation for this reaction, including state symbols.

[1]

ii)

Using oxidation numbers, state whether the reaction is a redox reaction.

[2]

[3 marks]

Question 3a

Silicon(IV) chloride is formed by the reaction of silicon and chlorine.

i)

Write an equation for this reaction.

[1]

ii)

Draw the structure of silicon(IV) chloride.

[1]

iii)

Explain why silicon chloride is a liquid at room temperature.

[1]

[3 marks]

Question 3b

The reaction of silicon(IV) chloride and water is very violent giving a white solid, silicon dioxide and steamy fumes of hydrogen chloride.

Tetrachloromethane gives no reaction with water.

Using your knowledge of electronic configuration, explain the difference between these two observations.

[3 marks]

Question 3c

i)

Write an equation which demonstrates the acidic property of silicon dioxide.

[1]

ii)

Explain why silicon dioxide cannot act as a base.

[1]

[2 marks]

Model Answers: Hard

1a

a) Equations, observations and resulting pH's are:

The reaction of sodium oxide with water

- $\text{Na}_2\text{O (s)} + \text{H}_2\text{O (l)} \rightarrow 2\text{NaOH (aq)}$; [1 mark]
- Vigorous / very exothermic reaction

AND

A colourless solution produced; [1 mark]

- pH in the range of 13-14; [1 mark]

The reaction of sulfur dioxide with water

- $\text{SO}_2 \text{ (g)} + \text{H}_2\text{O (l)} \rightarrow \text{H}_2\text{SO}_3 \text{ (aq)}$; [1 mark]
- A colourless solution produced; [1 mark]
- pH in the range of 1-4; [1 mark]

[Total: 6 marks]

- **Remember:** The acid-base character of the Period 3 oxides changes as you go from left to right:
-

Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_2 SO_3
Very strong alkali	Strong alkali	Amphoteric	Acidic	Strong acid	Strong acid

1b

b) The ionic equation for the reaction is:

- $\text{SiO}_2 + 2\text{OH}^- \rightarrow \text{SiO}_3^{2-} + \text{H}_2\text{O}$; [1 mark]

[Total: 1 mark]

- When sodium oxide, Na_2O reacts with water it forms sodium hydroxide, NaOH
- As hydrogen is present in the reactants, water is produced
- If a concentrated solution of NaOH is heated with silicon dioxide, SiO_2 , an acid-base reaction occurs producing Na_2SiO_3
 - $\text{SiO}_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SiO}_3 + \text{H}_2\text{O}$
- To write the ionic equation:
 - Split the ionic compounds into their ions
 - $\text{SiO}_2 + 2\text{Na}^+ + 2\text{OH}^- \rightarrow 2\text{Na}^+ + \text{SiO}_3^{2-} + \text{H}_2\text{O}$
 - Remove the spectator ions (Na^+)
 - $\text{SiO}_2 + 2\text{OH}^- \rightarrow \text{SiO}_3^{2-} + \text{H}_2\text{O}$

1c

c)

i) The term used to describe the behaviour of compound **A**

- Amphoteric; [1 mark]

ii) The reactions for compound **A** with potassium hydroxide and with nitric acid are:

- $\text{Al}_2\text{O}_3 (\text{s}) + 2\text{KOH} (\text{aq}) + 3\text{H}_2\text{O} (\text{l}) \rightarrow 2\text{KAl}(\text{OH})_4 (\text{aq})$; [1 mark]
- $\text{Al}_2\text{O}_3 (\text{s}) + 6\text{HNO}_3 (\text{aq}) \rightarrow 2\text{Al}(\text{NO}_3)_3 (\text{aq}) + 3\text{H}_2\text{O}$; [1 mark]

[Total: 3 marks]

- You will know that the oxide of aluminium is amphoteric in nature
 - This means that it can act as an acid or a base
- Therefore, you can assume Compound **A** is Al_2O_3
 - The only other insoluble oxide in Period 3 is SiO_2 , which will only react with an alkali as it is acidic in nature
- Sodium hydroxide is the most common chemical used for the reaction between Al_2O_3 and a base, but potassium hydroxide will also react with it forming a salt
 - **Careful:** The alkali must be hot and concentrated

2a

a) The acid-base character of the Period 3 oxides changes depending on the electronegativity of the Period 3 element because:

- The acidity / pH value / basicity is related to the difference in electronegativity between Period 3 element and oxide ion; [1 mark]

- The low electronegativity of Period 3 element causes X^+ and OH^- to form

AND

Therefore, the oxide is basic; [1 mark]

- The high electronegativity of Period 3 causes $X-O^-$ and H^+ to form

AND

Therefore, the oxide is acidic; [1 mark]

[Total: 3 marks]

- The more electropositive the Period 3 atom the more basic the oxide creating an ionic bond
- The more electronegative the Period 3 atom, the more acidic the oxide creating a covalent bond
- In an ionic Period 3 oxide
 - The oxygen can attract electron density towards itself in an ionic compound and electrons are transferred giving the O a negative charge and the Period 3 element a positive charge
 - The O^{2-} ion will therefore be able to form an OH^- ion which is basic
- In a covalent Period 3 oxide
 - There is little difference in electronegativity between the atoms in the oxide, therefore the H^+ ion is released causing the oxide to be acidic

2b

b)

i) The equation for the formation of phosphoric acid and the expected pH are:

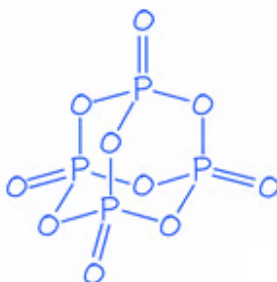
- $P_4O_{10}(s) + 6H_2O(l) \rightarrow 4H_3PO_4(aq)$; [1 mark]
- pH in the range of 1 - 2; [1 mark]

ii) The melting point of phosphorus(V) oxide is relatively low compared to most other Period 3 oxides because:

- Phosphorus(V) oxide / P_4O_{10} is a molecular compound; [1 mark]
- There are only van der Waals forces between molecules, which are overcome by heating; [1 mark]
- Ionic and giant covalent / macromolecular oxides have much higher melting points; [1 mark]

[Total: 5 marks]

- The formation of phosphoric acid from P_4O_{10} is a key equation to learn as it demonstrates the acidic character of a non-metal oxide
- The molecular structure of P_4O_{10} is:

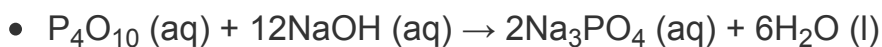


- Phosphorus(V) oxide, P_4O_{10} , has a melting point of 573 K which is lower than that of Na_2O , MgO and Al_2O_3 but higher than SO_2 and SO_3

2c

c)

i) The equation is:



OR



ii) Using oxidation numbers to determine if the reaction is redox:

- Sodium remains +1 in $NaOH$ and Na_3PO_4 ;

AND

Phosphorus remains +5 in both P_4O_{10} and Na_3PO_4 ; [1 mark]

- Therefore, the reaction is not a redox reaction; [1 mark]

[Total: 3 marks]

- When P_4O_{10} (aq) reacts with water in a solution it will form H_3PO_4 (aq) so the reaction between NaOH (aq) and H_3PO_4 (aq) is also a possible answer
- This is an example of an acid-base reaction and not a redox reaction
 - A redox reaction involves oxidation and reduction occurring
 - As the oxidation numbers are not changing, neither oxidation nor reduction is occurring

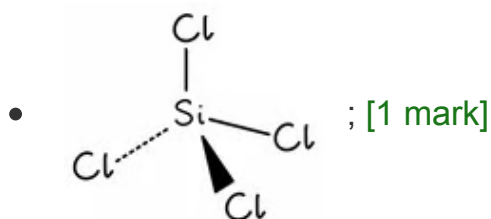
3a

a)

i) The equation for the reaction of silicon and chlorine to form silicon(IV) chloride is:

- $\text{Si} + 2\text{Cl}_2 \rightarrow \text{SiCl}_4$; [1 mark]

ii) The structure of silicon(IV) chloride is:



iii) Silicon(IV) chloride is a liquid at room temperature because:

- There are only van der Waals forces between SiCl_4 molecules
AND
Which require little energy to overcome; [1 mark]

[Total: 3 marks]

- To balance the equation
 - The only element that needs balancing is chlorine
 - $\text{Si} + \text{Cl}_2 \rightarrow \text{SiCl}_4$

- Therefore, the reaction is not a redox reaction; [1 mark]

[Total: 3 marks]

- When P_4O_{10} (aq) reacts with water in a solution it will form H_3PO_4 (aq) so the reaction between NaOH (aq) and H_3PO_4 (aq) is also a possible answer
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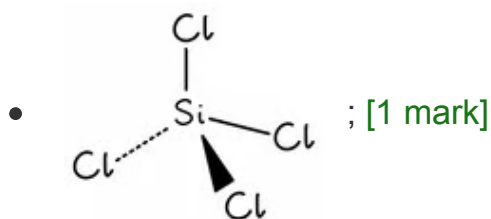
3a

a)

i) The equation for the reaction of silicon and chlorine to form silicon(IV) chloride is:

- $\text{Si} + 2\text{Cl}_2 \rightarrow \text{SiCl}_4$; [1 mark]

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- There are only van der Waals forces between SiCl_4 molecules
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Which require little energy to overcome; [1 mark]

[Total: 3 marks]

- To balance the equation
 - The only element that needs balancing is chlorine
 - $\text{Si} + \text{Cl}_2 \rightarrow \text{SiCl}_4$
 - $\text{Si} + 2\text{Cl}_2 \rightarrow \text{SiCl}_4$
- To draw the structure

- There are 4 electrons in the 3rd orbit and silicon will behave in a similar way to carbon
- You know that tetrachloromethane will have a tetrahedral structure, therefore silicon(IV) chloride will have the same molecular shape
- You can work out the number of electron pairs by
 - $\frac{\text{group number} + \text{bonding atoms}}{2} = \frac{4 + 4}{2} = 4$ electron pairs
 - All are involved in bonding, so the shape must be tetrahedral
- OR
- There are 4 outer electrons in each carbon atom and 7 outer electrons in each chlorine atom
- Therefore 32 outer electrons
 - 24 electrons from the chlorine atoms are in lone pairs (3 lone pairs on each chlorine atom, so 6×4)
 - This leaves 8 electrons which will all be involved in the four bonds, so no lone pairs around the central atom
 - 4 bonding pairs means a tetrahedral shape
- Silicon(IV) dioxide is a liquid because it is a simple molecule
 - It is also non-polar due to the fact that it is a symmetrical molecule

3b

b) The difference between these observations is due to:

- Silicon / Si atom has vacant 3d orbitals

OR

Carbon / C atom does not have vacant 3d orbitals; [1 mark]

- Oxygen can bond with silicon / Si atoms (before breaking Si-Cl bonds); [1 mark]
- A large amount of energy is released during this reaction which can not occur in the CCl_4 reaction

OR

There is a high activation energy requirement for CCl_4 and H_2O reaction; [1 mark]

[Total: 3 marks]

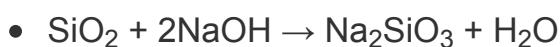
- This question relies less on your knowledge of Period 3 chlorides and more on your knowledge of atomic structure
- Think about the comparison, the only thing that is changing is the central atom, Si to C

- Carbon has the same number of outer electrons but is smaller
 - This creates two problems
 1. Chlorine atoms are bulky and will 'get in the way' of any incoming reactant creating repulsion
 2. There isn't a convenient empty orbital on the carbon that the oxygen lone pair can attach to
- Silicon is much bigger and also has vacant 3d orbitals meaning the oxygen can bond to the silicon before the need to break a silicon-chlorine bond
 - This makes the whole process energetically easier releasing energy
 - The overall violent reaction is
 - $\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$
 - This is a reaction you must know

3c

c)

i) An equation that demonstrates the acidic nature of silicon dioxide is:



OR



ii) Silicon dioxide does not behave as a base because:

- It does not contain oxide / O^{2-} ions; [1 mark]

[Total: 2 marks]

- Silicon dioxide is a giant covalent structure that can behave as a weak acid
 - When it reacts with an alkali, the alkali must be hot and concentrated
- It does not form oxide ions
 - Therefore, it cannot generate OH^- ions in water
- It can also act as a base in the blast furnace removing calcium oxide to form calcium silicate or slag